

Square Numbers — Answers

Final answers with a one-line reason. For full step-by-step methods, see the worked examples in the notes.

Objective

- Q1** (c) **196**. $196 = 14^2$, and even² is even. The others are 13^2 , 15^2 , 21^2 — all odd.
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- Q2** (b) **10**. Numbers between n^2 and $(n+1)^2 = 2n = 2 \times 5$.
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- Q3** (c) **24**. $4 = 2^2$, $9 = 3^2$, $100 = 10^2$; 24 is no whole number squared.
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- Q4** **4**. Only the last digit matters: $2^2 = 4$.
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- Q5** **False**. Squares never end in 2, 3, 7 or 8.
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- Q6** (b) **Both true, but R does not explain A**. $90000 = 300^2$, so A is true; the even-zeros rule is true but only necessary, not sufficient (it cannot by itself prove a number is a square).
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Short answer

- Q7** **6 zeros**. Trailing zeros double on squaring ($3 \rightarrow 6$).
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- Q8** **All four are not perfect squares**. 1057 ends in 7; 23453 ends in 3; 7928 ends in 8 (forbidden last digits); 222000 has 3 trailing zeros — an odd number of zeros.
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- Q9** **100, 121, 144**. These are 10^2 , 11^2 , 12^2 . (81 is below 90; 169 is above 150.)
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Long answer

- Q10** $\sqrt{36} = 6$. $36 - 1 - 3 - 5 - 7 - 9 - 11$ reaches 0 after 6 subtractions $\Rightarrow$ perfect square, root = 6.
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- Q11** $36^2 = 1296$. $36^2 = 35^2 + (35 + 36) = 1225 + 71$.
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- Q12** **Side = 13 m**. $13^2 = 169$; its units digit 9 is an allowed square-ending, which confirms 169 can be a perfect square.
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Total: 20 marks.